

ZA-GAS: phi-only vs phi-xi — Belo Horizonte Pilot Study

ZA-GAS Model Comparison — Thesis Working Report

Abstract

This report compares two specifications of the Zero-Augmented Generalised Autoregressive Score (ZA-GAS) model for precipitation time series. In the phi-only specification only the GB2 location parameter ϕ is time-varying; in the phi-xi specification both ϕ and the shape parameter ξ follow the GAS score-driven recursion. Models are evaluated at daily and monthly cadences on in-sample fit (log-likelihood, AIC, BIC) and out-of-sample forecast accuracy (CRPS, RMSE, MAD). Probabilistic calibration is assessed via PIT uniformity tests (Kolmogorov-Smirnov) and 95 % coverage tests (Kupiec, Christoffersen).

Scope Note

This is the first experiment in the thesis research. Models were fitted and evaluated only for the Belo Horizonte (BH) time series. Results for other Brazilian weather stations will be presented in subsequent experiments once the full spatial sweep is complete.

Model Descriptions

Framework: Zero-Augmented GAS

The observation equation is zero-augmented. Conditional on information F_{t-1} ,

$$p(y_t | F_{t-1}) = (1 - \pi_t) \mathbf{1}\{y_t = 0\} + \pi_t g(y_t; f_{t|t-1}, \theta) \mathbf{1}\{y_t > 0\}.$$

The positive component $g(\cdot)$ is GB2 with log-link parameters. The state vector $f_{t|t-1}$ contains the time-varying positive-part parameters: either $f_{t|t-1} = (\phi_{t|t-1})$ or $f_{t|t-1} = (\phi_{t|t-1}, \xi_{t|t-1})'$.

The GAS recursion follows the one-step-ahead prediction notation used in score-driven models:

$$f_{t+1|t} = \omega + \sum_{\ell \in L} A_\ell s_{t-\ell+1} + \sum_{\ell \in L} B_\ell f_{t-\ell+1|t-\ell}.$$

Here $s_t = \frac{\partial}{\partial f_{t|t-1}} \log p(y_t | F_{t-1})$ is the Fisher-information-scaled score, $\frac{\partial}{\partial f_{t|t-1}} = \frac{\partial}{\partial f_{t|t-1}} \log p(y_t | F_{t-1})$. The score is set to zero for exact-zero observations because those observations identify the zero probability rather than the positive GB2 density.

For the non-seasonal GAS(1,1) special case this reduces to $f_{t+1|t} = \omega + A s_t + B f_{t|t-1}$.

Pi Dynamics

The non-zero probability is $\pi_t = \Lambda(\eta_t)$, where $\Lambda(x) = 1/(1 + \exp(-x))$ is the logistic link.

The implemented AR-logistic recursion is $\eta_t = \omega_0 + \rho \eta_{t-1} + \sum_{\ell \in L} \omega_{y,\ell} y_{t-\ell}$.

The same seasonal lag set L is used for the GAS recursion and the π_t dynamics, so the zero/non-zero occurrence process and the positive-part intensity process can both react to short-memory and calendar-seasonal precipitation information.

Seasonal Lag Sets

Daily cadence: $L = \{1, 2, 3, 364, 365, 366, 367\}$ - seven lags covering +/-1 day around the annual cycle

Monthly cadence: $L = \{1, 2, 3, 11, 12, 13\}$ - six lags covering +/-1 month around the annual cycle

Model 1 - ϕ -only

Only ϕ is time-varying; ξ is kept at its estimated constant value throughout the sample. This is the parsimonious baseline:

it captures location shifts in positive-part precipitation while holding the spread/shape channel fixed.

Model 2 - ϕ, ξ

Both ϕ and ξ follow the GAS update. This allows both the positive-part scale/location and the GB2 shape channel to react dynamically to new observations, potentially improving calibration in regimes where rainfall intensity and dispersion change together.

Parameter Counts

Daily ['phi']: 28 estimated parameters (16 GAS dynamics, 3 static positive-part, 9 pi-dynamics)

Daily ['phi', 'xi']: 43 estimated parameters (32 GAS dynamics, 2 static positive-part, 9 pi-dynamics)

Monthly ['phi']: 25 estimated parameters (14 GAS dynamics, 3 static positive-part, 8 pi-dynamics)

Monthly ['phi', 'xi']: 38 estimated parameters (28 GAS dynamics, 2 static positive-part, 8 pi-dynamics)

Estimation

Parameters are estimated by maximising the full log-likelihood using L-BFGS-B with box constraints. Standard errors are Wald-type from the optimizer's inverse Hessian approximation. Confidence intervals are at the 95% level.

Series Diagnostics

Section Overview

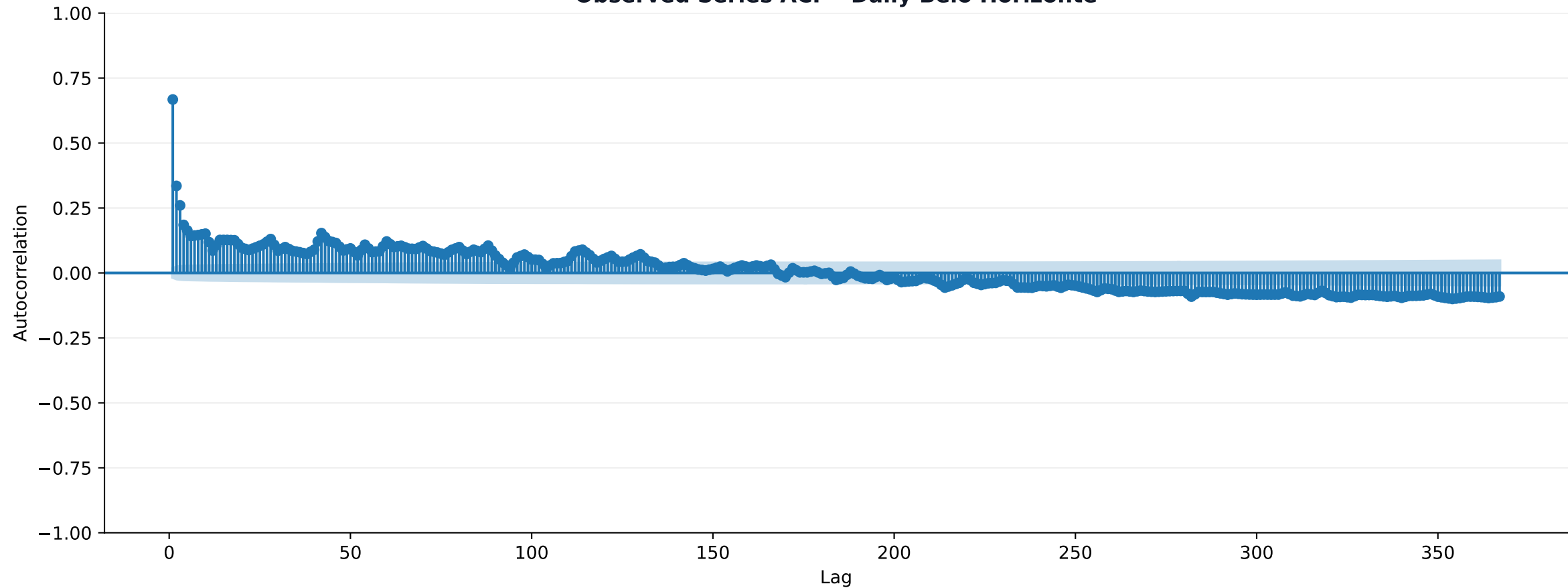
This section describes the observed precipitation series before any model-specific diagnostics are shown. The statistics and ACF charts are about the data series itself, not about a fitted model.

Observed Series Descriptive Statistics

Series	sample	n	min	max	mean	std	prop_zero	q01	q05	q25	q50	q75	q90	q95	q99
Daily Belo Horizonte	IS	3650.000	0.000	171.800	4.060	11.703	0.720	0.000	0.000	0.000	0.000	0.600	14.110	26.110	53.961
Daily Belo Horizonte	OOS	730.000	0.000	92.800	4.217	11.165	0.708	0.000	0.000	0.000	0.000	0.575	15.000	29.255	51.678
Daily Belo Horizonte	Full	4380.000	0.000	171.800	4.086	11.614	0.718	0.000	0.000	0.000	0.000	0.600	14.200	26.505	53.321
Monthly Belo Horizonte	IS	216.000	0.000	934.700	130.507	148.715	0.093	0.000	0.000	14.300	82.150	212.425	332.250	399.325	595.720
Monthly Belo Horizonte	OOS	24.000	0.000	410.100	126.504	135.759	0.208	0.000	0.000	4.675	91.700	234.025	328.550	345.585	395.265
Monthly Belo Horizonte	Full	240.000	0.000	934.700	130.106	147.208	0.104	0.000	0.000	11.975	82.150	214.050	334.930	397.145	586.792

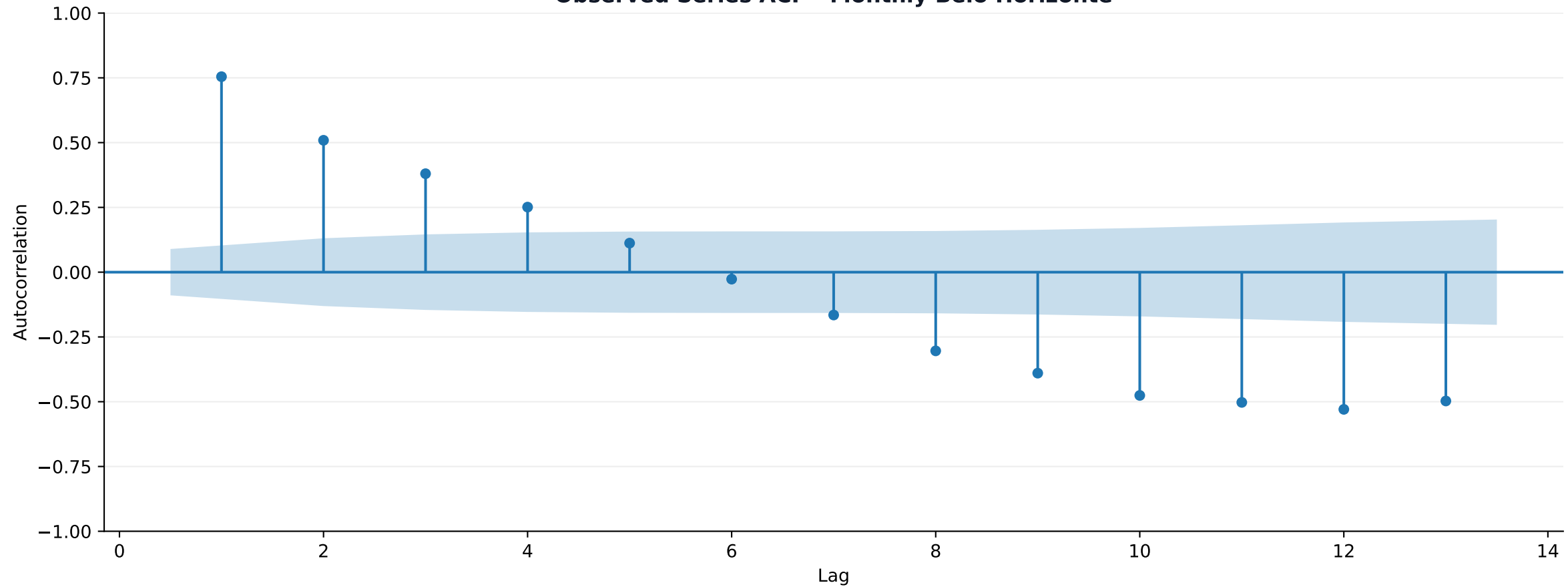
Note: Statistics are computed for the raw precipitation series. Full = IS and OOS concatenated.

Observed Series ACF - Daily Belo Horizonte



ACF of observed precipitation, lags 1-367; shaded bands are approximate 95% confidence intervals.

Observed Series ACF - Monthly Belo Horizonte



ACF of observed precipitation, lags 1-13; shaded bands are approximate 95% confidence intervals.

Per-Model Diagnostics

Section Overview

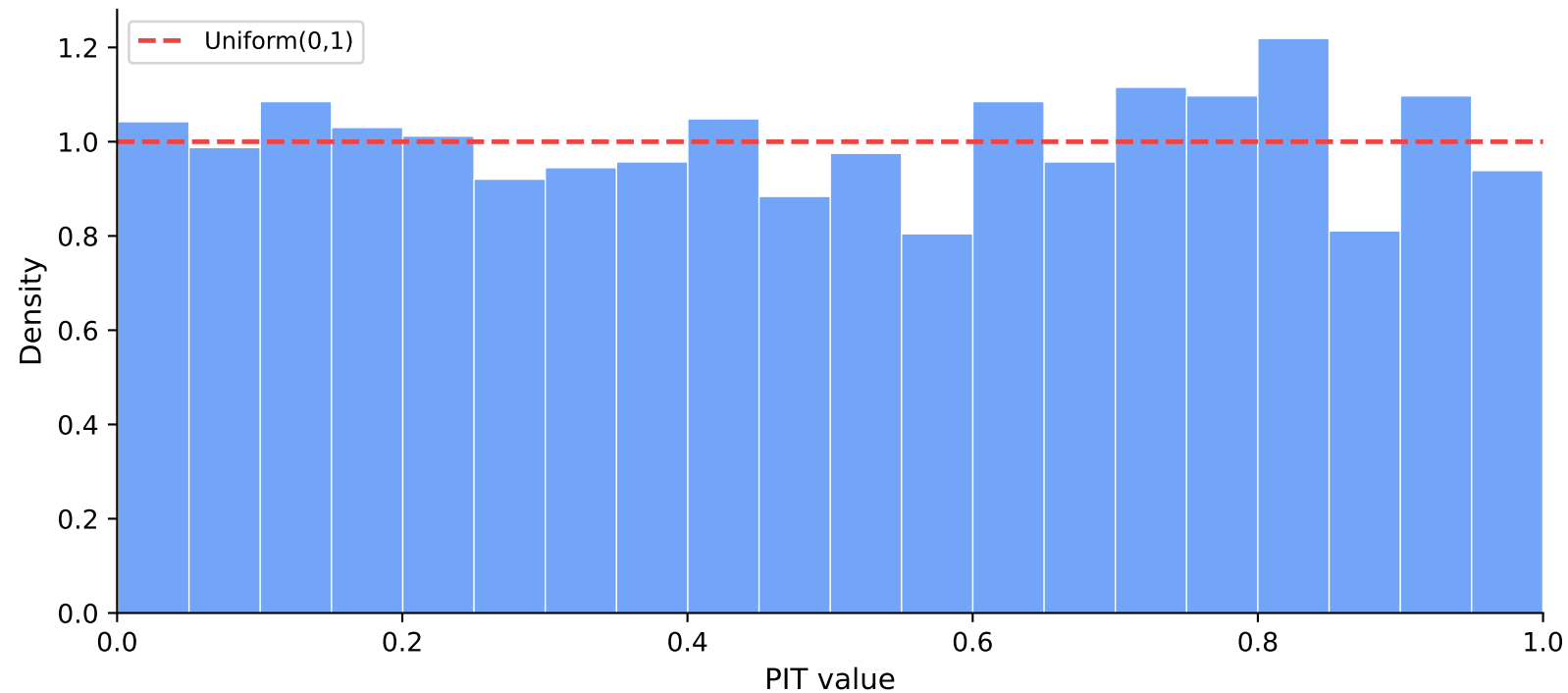
The following pages present individual diagnostic charts and tables for each fitted model: PIT histogram (calibration), quantile residual ACF (serial independence), KS test statistics, and estimated parameters. All diagnostic charts use only in-sample observations. Cross-model comparisons are deferred to the Comparison Tables section.

Diagnostics: Daily ϕ -only

Scope

This section presents in-sample diagnostic charts and tables for the Daily ϕ -only model. These are model-specific checks of probabilistic calibration (PIT) and residual autocorrelation (ACF). Cross-model comparisons are in the Comparison Tables section.

In-Sample PIT Histogram — Daily ϕ -only



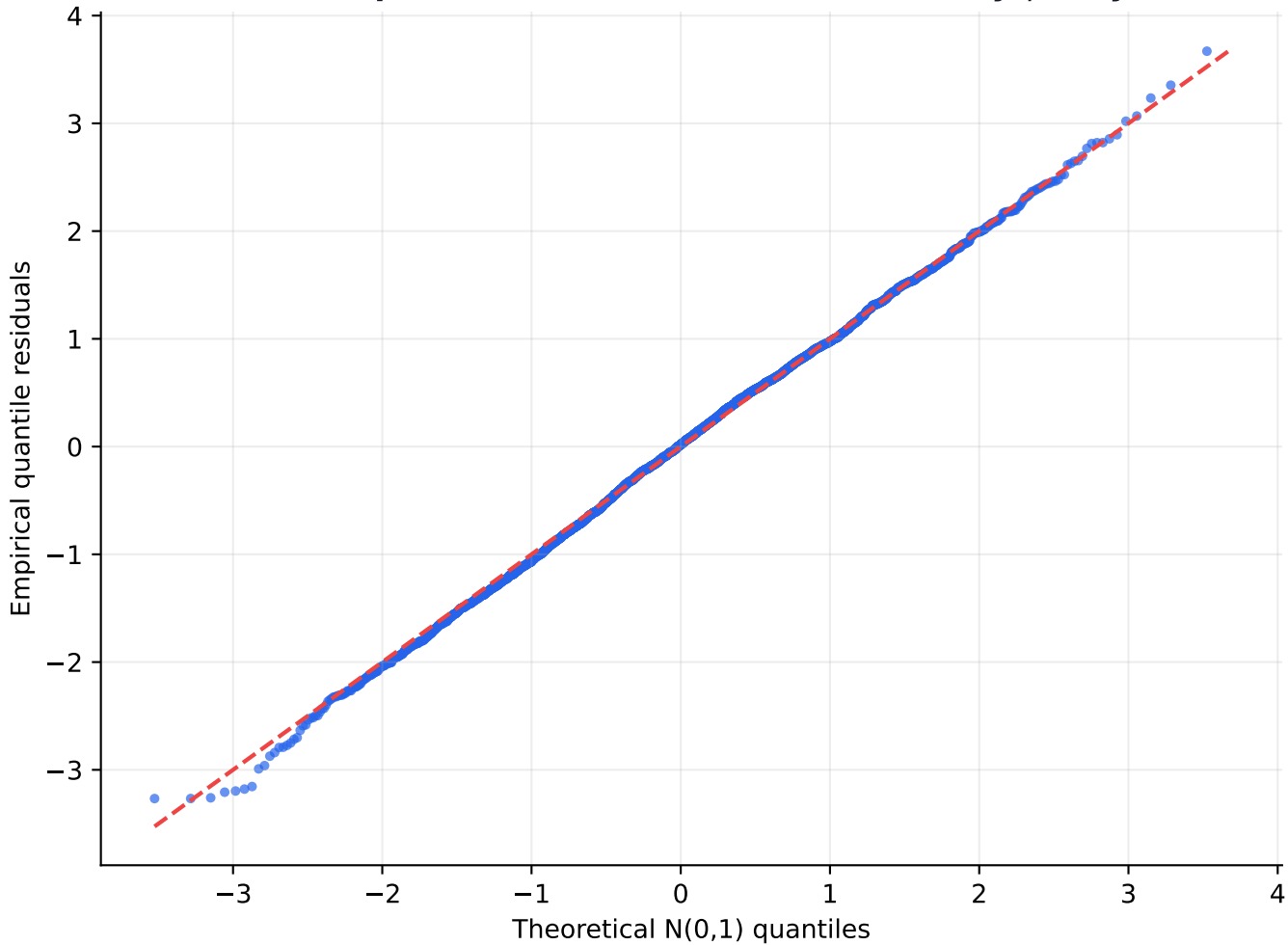
Bars close to height 1 (the red dashed line) indicate good probabilistic calibration.

PIT Kolmogorov-Smirnov Test — Daily ϕ -only

sample	n	ks_stat	pvalue	reject_5pct
IS	3285.000	0.017	0.328	0.000
OOS	731.000	0.025	0.745	0.000
Full	4016.000	0.013	0.477	0.000

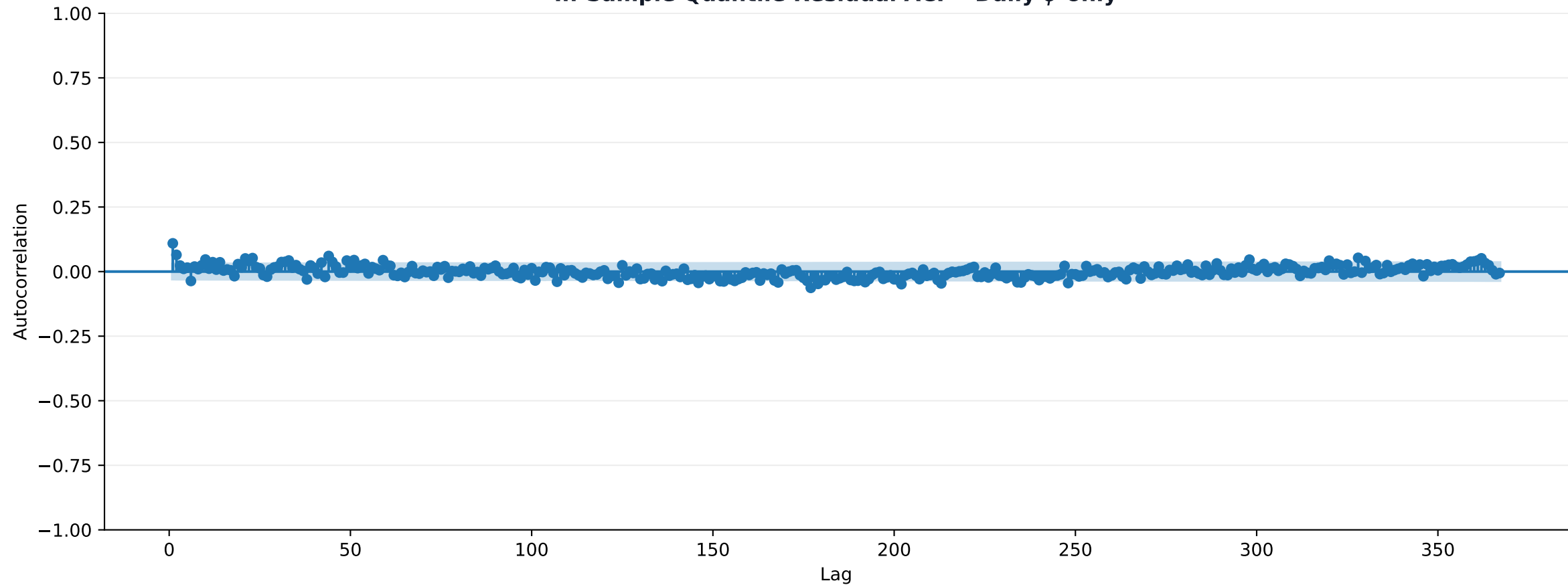
Note: KS test of PIT values against Uniform(0, 1). A well-calibrated model has p-value > 0.05 and ks_stat close to zero. IS = in-sample; OOS = out-of-sample; Full = concatenated sample.

In-Sample Quantile Residual QQ Plot - Daily ϕ -only



Points close to the dashed 45-degree line indicate approximately standard-normal quantile residuals.

In-Sample Quantile Residual ACF - Daily ϕ -only



One model only. Lags 1-367; shaded bands are approximate 95% confidence intervals under white-noise residuals.

Jarque-Bera Test for Quantile Residuals - Daily ϕ -only

sample	n	skewness	kurtosis	JB_stat	pvalue	reject_5pct
IS	3283.000	-0.062	2.943	2.551	0.279	0.000

Note: Jarque-Bera tests whether in-sample quantile residuals are compatible with a Gaussian distribution. Red cells indicate rejection at 5%.

Estimated Parameters — Daily ϕ -only

parameter	block	estimate	std_error	ci_lower_95	ci_upper_95
omega_phi	gas	0.7520	15.3518	-29.3370	30.8410
f0_phi	gas	2.9685	10.3172	-17.2528	23.1897
A_phi_1	gas	0.1467	1.6831	-3.1522	3.4456
A_phi_2	gas	0.0705	3.4295	-6.6511	6.7921
A_phi_3	gas	-0.0904	5.2673	-10.4141	10.2333
A_phi_364	gas	-0.0244	1.2954	-2.5633	2.5145
A_phi_365	gas	0.0680	3.2966	-6.3932	6.5292
A_phi_366	gas	0.0198	2.3063	-4.5005	4.5402
A_phi_367	gas	0.0307	1.3329	-2.5818	2.6431
B_phi_1	gas	0.2386	13.1448	-25.5248	26.0019
B_phi_2	gas	0.3688	13.3518	-25.8002	26.5378
B_phi_3	gas	0.2709	3.2488	-6.0965	6.6384
B_phi_364	gas	0.2756	1.9272	-3.5015	4.0528
B_phi_365	gas	-0.2428	15.7955	-31.2014	30.7159
B_phi_366	gas	-0.3203	10.3174	-20.5419	19.9014
B_phi_367	gas	0.2866	4.2760	-8.0942	8.6675
xi	positive_static	-0.1696	2.8394	-5.7348	5.3956
gamma	positive_static	0.1000	1.0000	-1.8600	2.0600
zeta	positive_static	3.1932	33.0527	-61.5889	67.9754
omega0	pi	-1.2557	2.2446	-5.6550	3.1436
rho	pi	0.2847	1.2475	-2.1602	2.7297
omega_y_1	pi	0.0556	0.6702	-1.2580	1.3692
omega_y_2	pi	0.0205	0.5940	-1.1437	1.1847
omega_y_3	pi	0.0130	0.8623	-1.6771	1.7032
omega_y_364	pi	0.0141	0.7219	-1.4009	1.4290
omega_y_365	pi	0.0059	0.8339	-1.6285	1.6403
omega_y_366	pi	0.0080	0.4887	-0.9498	0.9657
omega_y_367	pi	0.0082	0.7096	-1.3825	1.3990

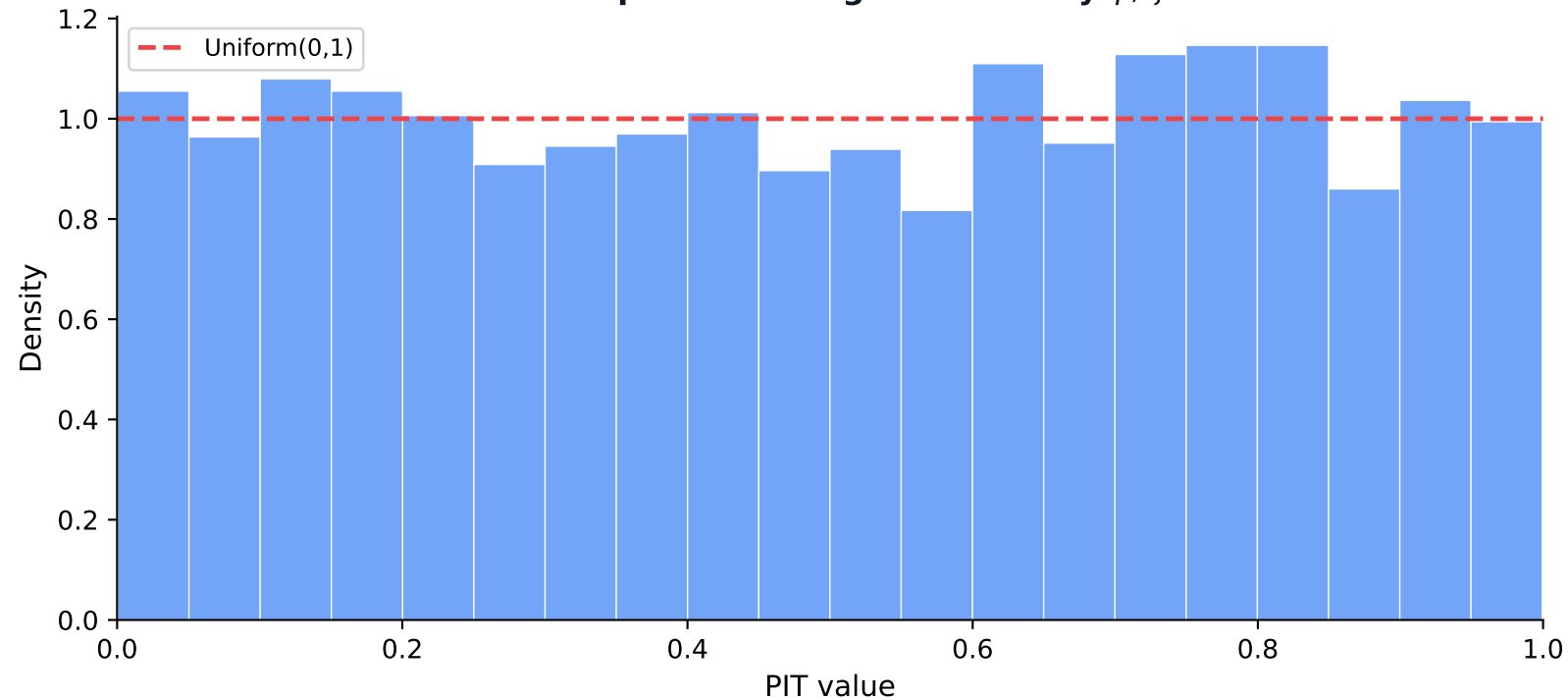
Note: Estimates from the optimizer used for the cached model. Standard errors and 95 % Wald CIs from the inverse Hessian; interpret with caution near optimizer bounds.

Diagnostics: Daily ϕ, ξ

Scope

This section presents in-sample diagnostic charts and tables for the Daily ϕ, ξ model. These are model-specific checks of probabilistic calibration (PIT) and residual autocorrelation (ACF). Cross-model comparisons are in the Comparison Tables section.

In-Sample PIT Histogram — Daily ϕ, ξ



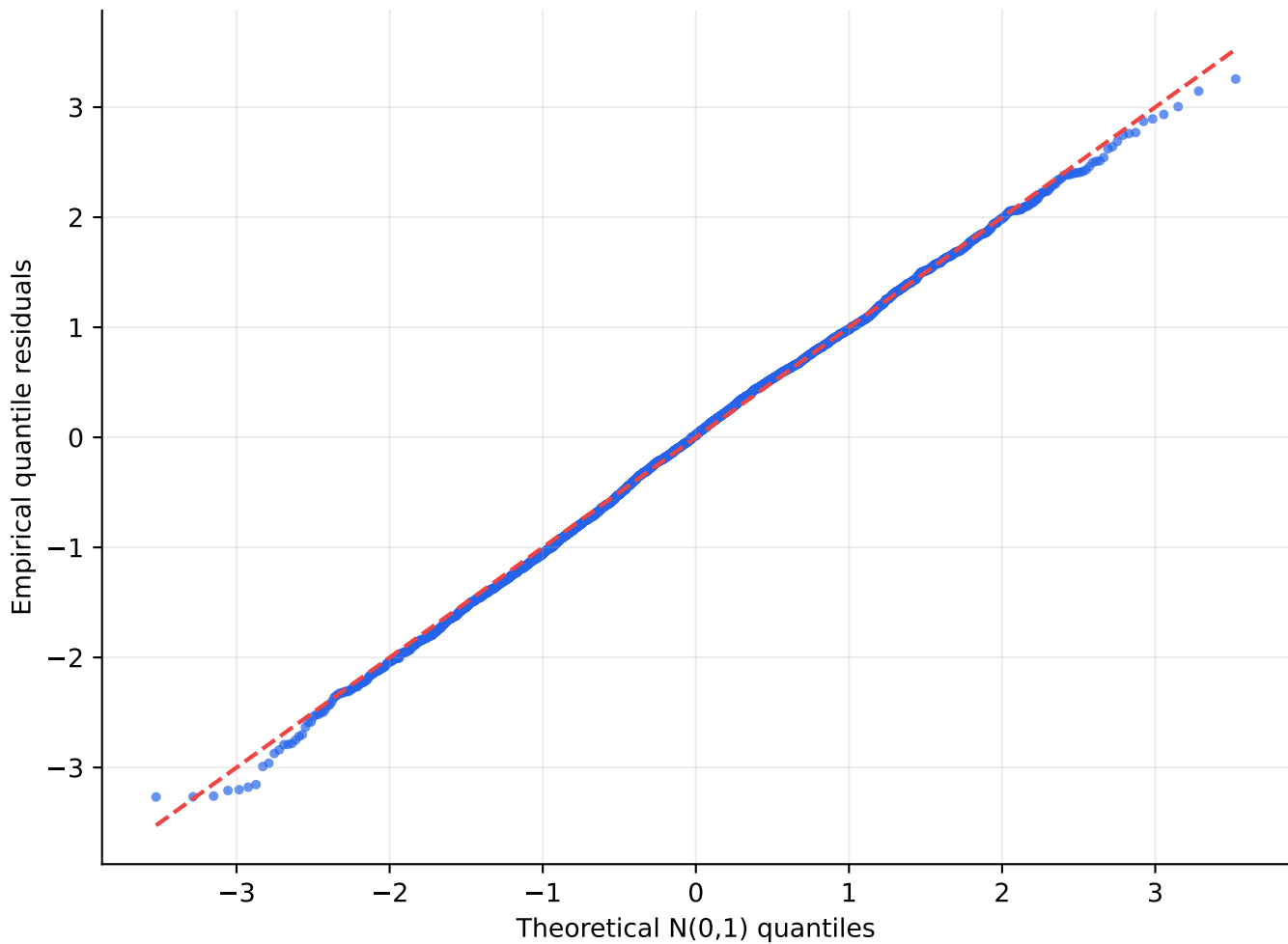
Bars close to height 1 (the red dashed line) indicate good probabilistic calibration.

PIT Kolmogorov-Smirnov Test — Daily ϕ, ξ

sample	n	ks_stat	pvalue	reject_5pct
IS	3285.000	0.020	0.130	0.000
OOS	731.000	0.024	0.784	0.000
Full	4016.000	0.016	0.278	0.000

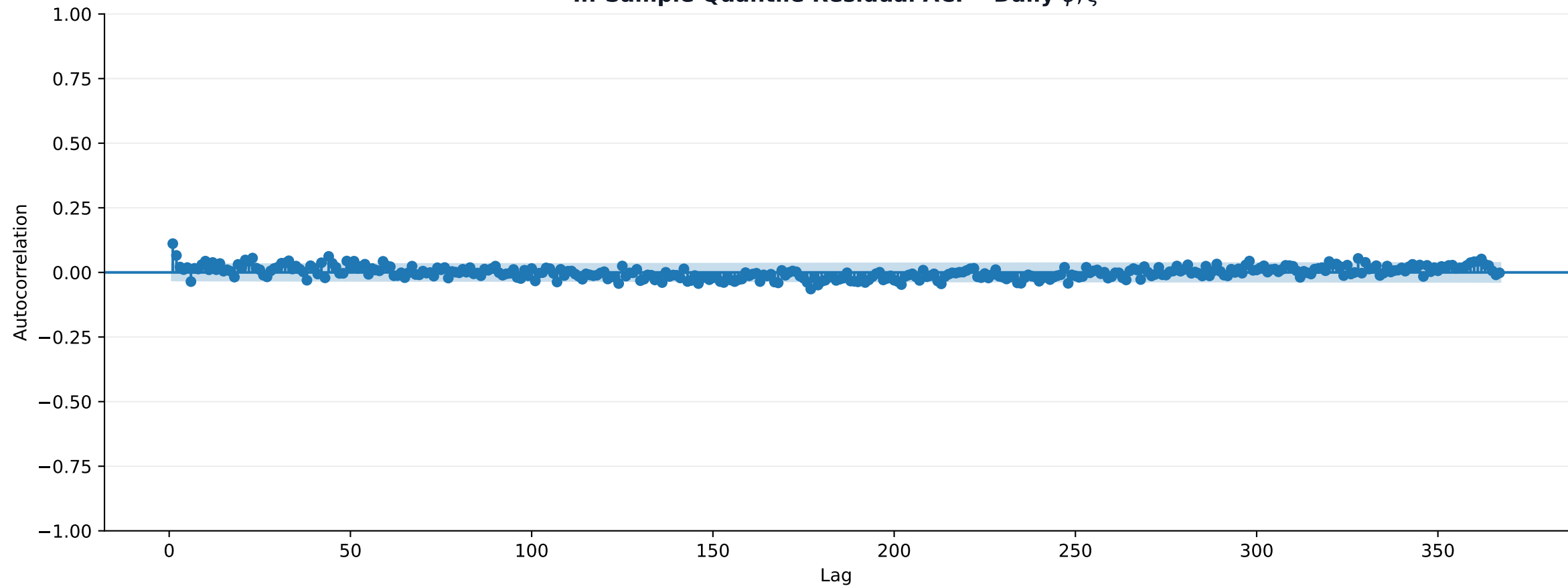
Note: KS test of PIT values against Uniform(0, 1). A well-calibrated model has p-value > 0.05 and ks_stat close to zero. IS = in-sample; OOS = out-of-sample; Full = concatenated sample.

In-Sample Quantile Residual QQ Plot - Daily ϕ, ξ



Points close to the dashed 45-degree line indicate approximately standard-normal quantile residuals.

In-Sample Quantile Residual ACF - Daily ϕ, ξ



One model only. Lags 1-367; shaded bands are approximate 95% confidence intervals under white-noise residuals.

Jarque-Bera Test for Quantile Residuals - Daily ϕ, ξ

sample	n	skewness	kurtosis	JB_stat	pvalue	reject_5pct
IS	3283.000	-0.087	2.885	5.918	0.052	0.000

Note: Jarque-Bera tests whether in-sample quantile residuals are compatible with a Gaussian distribution. Red cells indicate rejection at 5%.

parameter	block	estimate	std error	ci lower 95	ci upper 95
omega_phi	gas	0.8594	2.2849	-3.6188	5.3377
f0_phi	gas	2.9637	1.5350	-0.0448	5.9722
A_phi_1	gas	0.1403	1.1609	-2.1350	2.4156
A_phi_2	gas	0.0754	1.6335	-3.1261	3.2770
A_phi_3	gas	-0.1171	2.5939	-5.2010	4.9667
A_phi_364	gas	-0.0478	1.3321	-2.6586	2.5631
A_phi_365	gas	0.0752	1.8010	-3.4546	3.6050
A_phi_366	gas	0.0540	2.2904	-4.4350	4.5431
A_phi_367	gas	0.0506	3.5015	-6.8123	6.9134
B_phi_1	gas	0.1935	1.6770	-3.0933	3.4804
B_phi_2	gas	0.2991	1.6380	-2.9114	3.5096
B_phi_3	gas	0.3613	3.0364	-5.5899	6.3126
B_phi_364	gas	-0.0048	1.5975	-3.1359	3.1263
B_phi_365	gas	-0.0416	1.9311	-3.8265	3.7433
B_phi_366	gas	0.0101	2.4365	-4.7654	4.7855
B_phi_367	gas	0.0311	2.5378	-4.9429	5.0051
omega_xi	gas	0.0486	7.3349	-14.3274	14.4247
f0_xi	gas	0.3003	2.2696	-4.1481	4.7487
A_xi_1	gas	-0.0045	2.4086	-4.7252	4.7163
A_xi_2	gas	-0.0040	1.5077	-2.9590	2.9511
A_xi_3	gas	0.0245	2.8556	-5.5724	5.6214
A_xi_364	gas	0.0164	1.4808	-2.8859	2.9187
A_xi_365	gas	0.0582	3.3396	-6.4873	6.6037
A_xi_366	gas	-0.0600	1.1523	-2.3185	2.1985
A_xi_367	gas	-0.1555	2.3216	-4.7058	4.3949
B_xi_1	gas	-0.1009	2.4154	-4.8351	4.6333
B_xi_2	gas	-0.7496	5.0325	-10.6131	9.1138
B_xi_3	gas	0.2178	3.4357	-6.5161	6.9517
B_xi_364	gas	0.0477	2.9118	-5.6594	5.7548
B_xi_365	gas	-0.1371	3.6641	-7.3186	7.0444
B_xi_366	gas	-0.0116	3.4151	-6.7051	6.6818
B_xi_367	gas	-0.1944	1.6384	-3.4057	3.0169
gamma	positive_static	0.1810	3.2802	-6.2480	6.6100
zeta	positive_static	2.8006	4.2559	-5.5408	11.1420
omega0	pi	-1.2563	1.9764	-5.1300	2.6174
rho	pi	0.2851	1.1940	-2.0551	2.6254
omega_y_1	pi	0.0558	0.7207	-1.3567	1.4683
omega_y_2	pi	0.0204	0.5935	-1.1427	1.1836
omega_y_3	pi	0.0132	0.9809	-1.9093	1.9357
omega_y_364	pi	0.0139	0.8359	-1.6244	1.6522
omega_y_365	pi	0.0060	0.9094	-1.7764	1.7885
omega_y_366	pi	0.0078	0.9514	-1.8570	1.8725
omega_y_367	pi	0.0084	0.6807	-1.3258	1.3426

Estimated Parameters — Daily ϕ, ξ

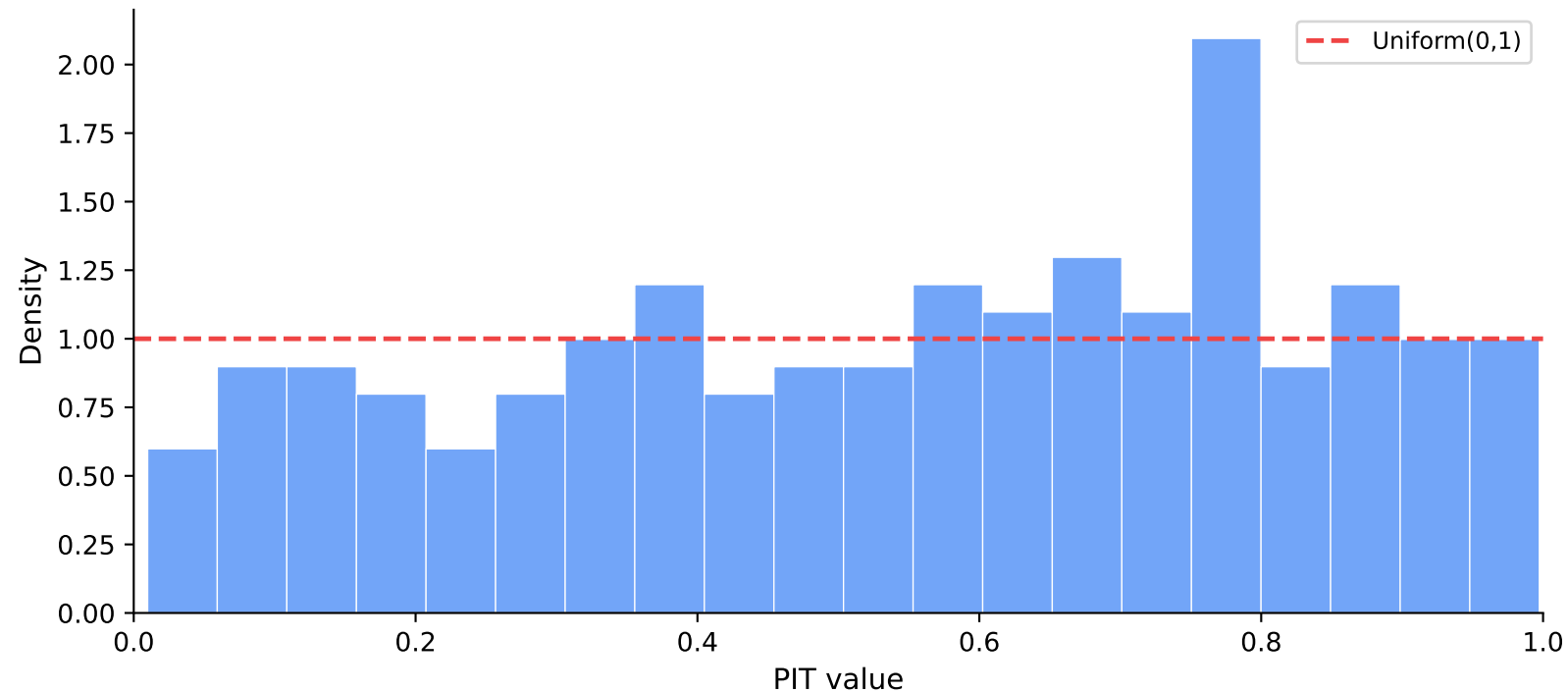
Note: Estimates from the optimizer used for the cached model. Standard errors and 95 % Wald CIs from the inverse Hessian; interpret with caution near optimizer bounds.

Diagnostics: Monthly ϕ -only

Scope

This section presents in-sample diagnostic charts and tables for the Monthly ϕ -only model. These are model-specific checks of probabilistic calibration (PIT) and residual autocorrelation (ACF). Cross-model comparisons are in the Comparison Tables section.

In-Sample PIT Histogram — Monthly ϕ -only



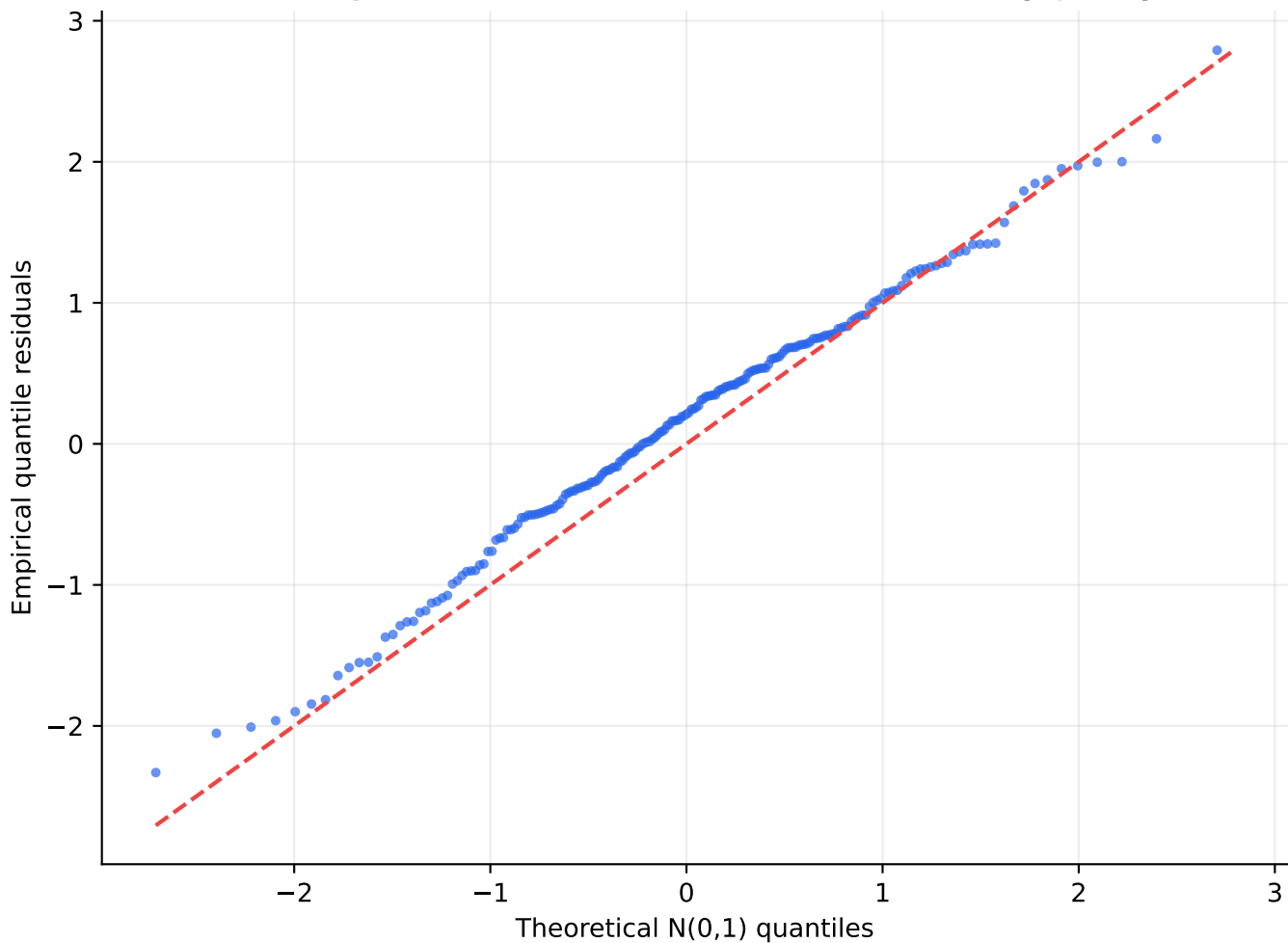
Bars close to height 1 (the red dashed line) indicate good probabilistic calibration.

PIT Kolmogorov-Smirnov Test — Monthly ϕ -only

sample	n	ks_stat	pvalue	reject_5pct
IS	203.000	0.097	0.042	1.000
OOS	24.000	0.138	0.700	0.000
Full	227.000	0.098	0.023	1.000

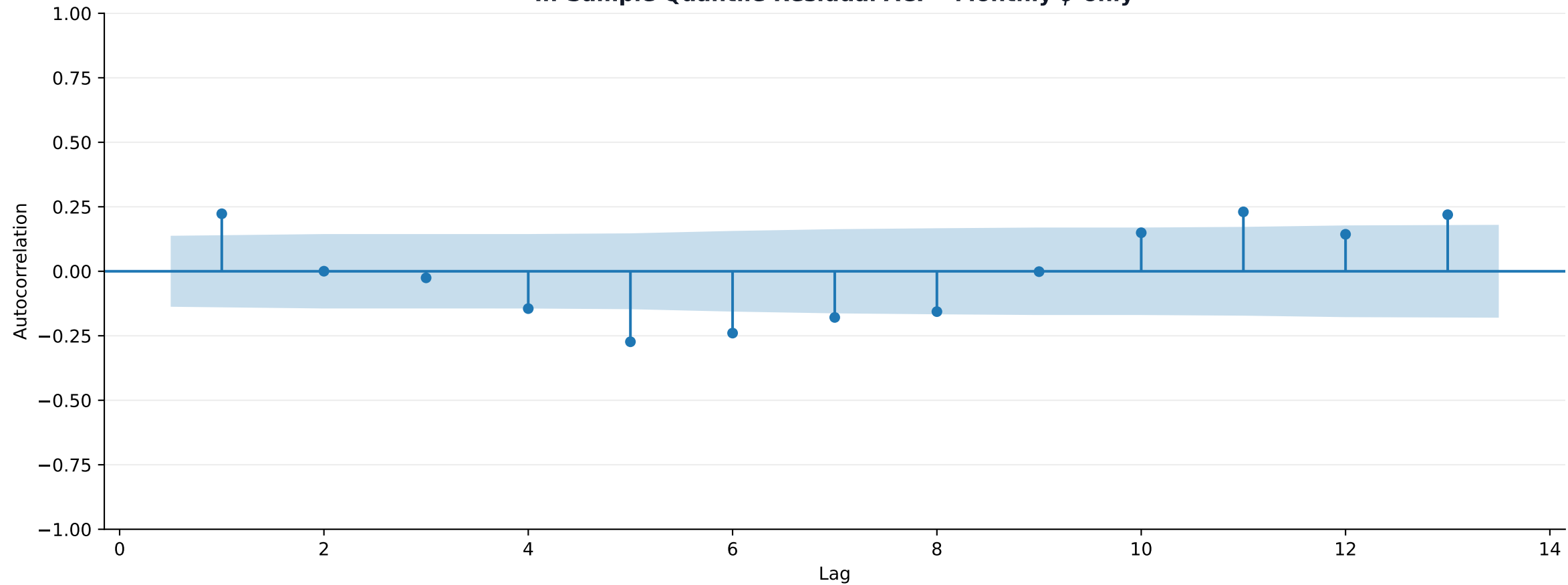
Note: KS test of PIT values against Uniform(0, 1). A well-calibrated model has p-value > 0.05 and ks_stat close to zero. IS = in-sample; OOS = out-of-sample; Full = concatenated sample.

In-Sample Quantile Residual QQ Plot - Monthly ϕ -only



Points close to the dashed 45-degree line indicate approximately standard-normal quantile residuals.

In-Sample Quantile Residual ACF - Monthly ϕ -only



One model only. Lags 1-13; shaded bands are approximate 95% confidence intervals under white-noise residuals.

Jarque-Bera Test for Quantile Residuals - Monthly ϕ -only

sample	n	skewness	kurtosis	JB_stat	pvalue	reject_5pct
IS	203,000	-0.193	2.979	1.265	0.531	0.000

Note: Jarque-Bera tests whether in-sample quantile residuals are compatible with a Gaussian distribution. Red cells indicate rejection at 5%.

Estimated Parameters — Monthly ϕ -only

parameter	block	estimate	std_error	ci_lower_95	ci_upper_95
omega_phi	gas	0.5592	0.0878	0.3871	0.7312
f0_phi	gas	10.1803	0.5218	9.1576	11.2030
A_phi_1	gas	0.1619	0.0295	0.1041	0.2197
A_phi_2	gas	0.0855	0.0343	0.0183	0.1528
A_phi_3	gas	-0.0064	0.0217	-0.0489	0.0362
A_phi_11	gas	0.1436	0.0160	0.1123	0.1749
A_phi_12	gas	0.3765	0.0369	0.3041	0.4488
A_phi_13	gas	0.2234	0.0458	0.1337	0.3131
B_phi_1	gas	-1.0246	0.0307	-1.0848	-0.9645
B_phi_2	gas	0.0257	0.0289	-0.0309	0.0823
B_phi_3	gas	0.0768	0.0312	0.0157	0.1379
B_phi_11	gas	0.1186	0.0310	0.0579	0.1794
B_phi_12	gas	0.9121	0.0377	0.8381	0.9861
B_phi_13	gas	0.8202	0.0358	0.7500	0.8905
xi	positive_static	-0.3907	0.2251	-0.8319	0.0505
gamma	positive_static	-0.3843	0.1634	-0.7045	-0.0641
zeta	positive_static	7.3490	0.3496	6.6638	8.0343
omega0	pi	-0.5405	0.3560	-1.2383	0.1574
rho	pi	0.0213	0.0407	-0.0584	0.1010
omega_y_1	pi	0.0078	0.0053	-0.0026	0.0182
omega_y_2	pi	0.0089	0.0042	0.0006	0.0172
omega_y_3	pi	0.0045	0.0022	0.0002	0.0089
omega_y_11	pi	0.0179	0.0077	0.0027	0.0330
omega_y_12	pi	0.0051	0.0087	-0.0119	0.0221
omega_y_13	pi	0.0048	0.0081	-0.0111	0.0206

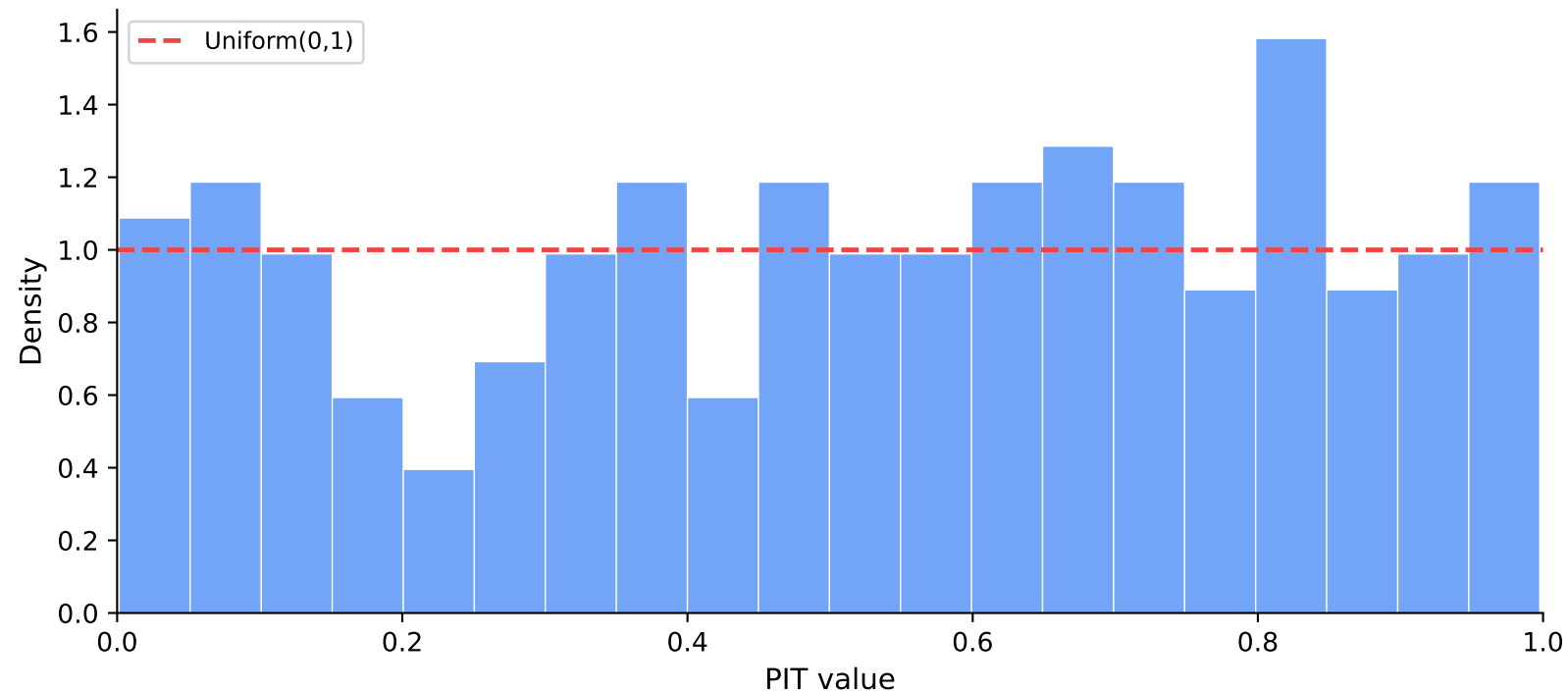
Note: Estimates from the optimizer used for the cached model. Standard errors and 95 % Wald CIs from the inverse Hessian; interpret with caution near optimizer bounds.

Diagnostics: Monthly ϕ, ξ

Scope

This section presents in-sample diagnostic charts and tables for the Monthly ϕ, ξ model. These are model-specific checks of probabilistic calibration (PIT) and residual autocorrelation (ACF). Cross-model comparisons are in the Comparison Tables section.

In-Sample PIT Histogram — Monthly ϕ, ξ



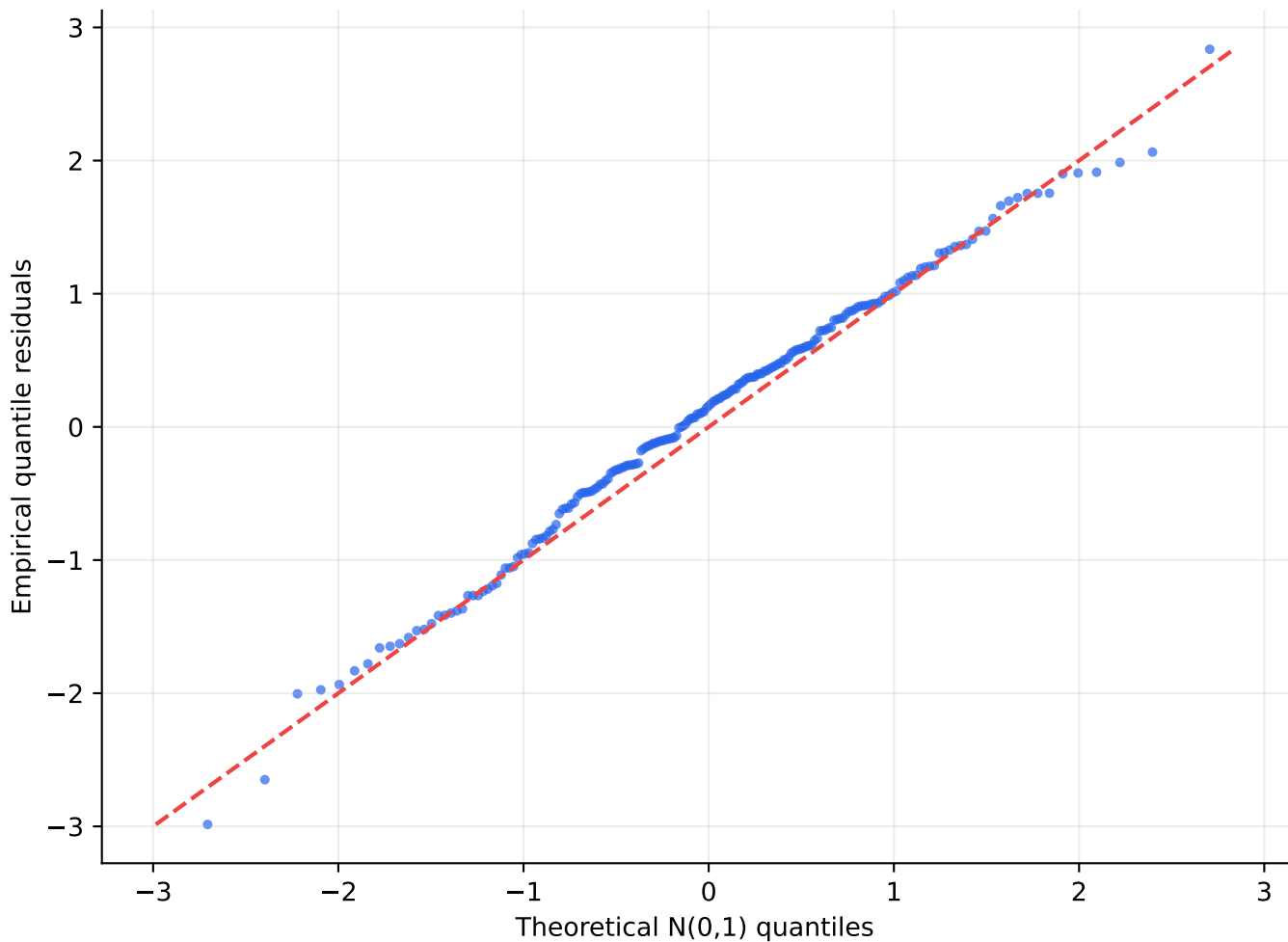
Bars close to height 1 (the red dashed line) indicate good probabilistic calibration.

PIT Kolmogorov-Smirnov Test — Monthly ϕ, ξ

sample	n	ks_stat	pvalue	reject_5pct
IS	203.000	0.070	0.255	0.000
OOS	24.000	0.249	0.085	0.000
Full	227.000	0.069	0.219	0.000

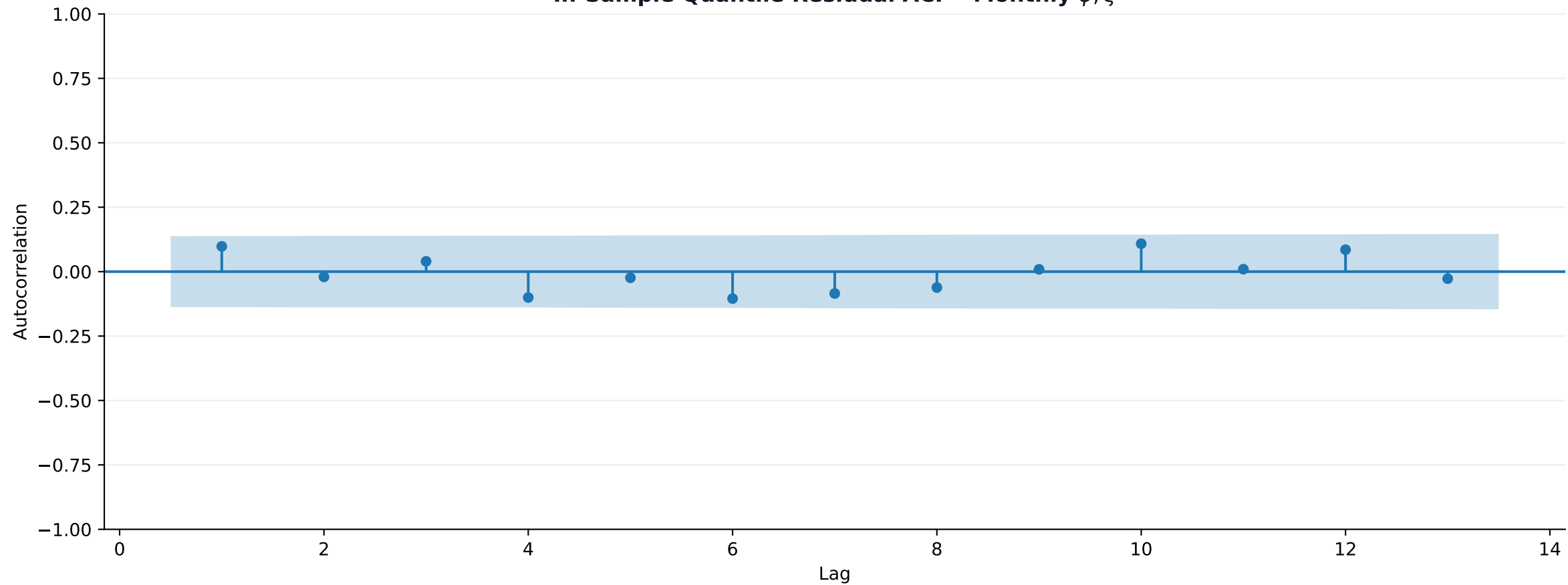
Note: KS test of PIT values against Uniform(0, 1). A well-calibrated model has p-value > 0.05 and ks_stat close to zero. IS = in-sample; OOS = out-of-sample; Full = concatenated sample.

In-Sample Quantile Residual QQ Plot - Monthly ϕ, ξ



Points close to the dashed 45-degree line indicate approximately standard-normal quantile residuals.

In-Sample Quantile Residual ACF - Monthly ϕ, ξ



One model only. Lags 1-13; shaded bands are approximate 95% confidence intervals under white-noise residuals.

Jarque-Bera Test for Quantile Residuals - Monthly ϕ, ξ

sample	n	skewness	kurtosis	JB_stat	pvalue	reject_5pct
IS	203,000	-0.246	3.003	2.052	0.358	0.000

Note: Jarque-Bera tests whether in-sample quantile residuals are compatible with a Gaussian distribution. Red cells indicate rejection at 5%.

parameter	block	estimate	std_error	ci_lower_95	ci_upper_95
omega_phi	gas	0.9038	0.6389	-0.3484	2.1560
f0_phi	gas	9.8781	0.2423	9.4031	10.3531
A_phi_1	gas	0.1287	0.0451	0.0403	0.2170
A_phi_2	gas	0.3903	0.0584	0.2759	0.5047
A_phi_3	gas	0.4727	0.0821	0.3117	0.6337
A_phi_11	gas	0.2683	0.0764	0.1186	0.4180
A_phi_12	gas	-0.1435	0.0484	-0.2384	-0.0486
A_phi_13	gas	-0.1681	0.0960	-0.3562	0.0200
B_phi_1	gas	0.0924	0.0705	-0.0457	0.2305
B_phi_2	gas	-0.5013	0.0769	-0.6520	-0.3505
B_phi_3	gas	0.3382	0.0426	0.2547	0.4216
B_phi_11	gas	0.3826	0.0359	0.3122	0.4529
B_phi_12	gas	0.2051	0.0595	0.0885	0.3216
B_phi_13	gas	0.3945	0.0537	0.2893	0.4997
omega_xi	gas	-0.1722	0.0288	-0.2287	-0.1157
f0_xi	gas	0.7362	0.0495	0.6392	0.8332
A_xi_1	gas	-0.1112	0.0545	-0.2180	-0.0043
A_xi_2	gas	-0.3695	0.0405	-0.4488	-0.2901
A_xi_3	gas	-0.6729	0.0409	-0.7532	-0.5927
A_xi_11	gas	0.1969	0.0662	0.0671	0.3268
A_xi_12	gas	0.2906	0.0530	0.1868	0.3944
A_xi_13	gas	0.4639	0.0509	0.3643	0.5636
B_xi_1	gas	-0.2183	0.0717	-0.3589	-0.0778
B_xi_2	gas	0.0386	0.0735	-0.1055	0.1827
B_xi_3	gas	0.1959	0.0501	0.0977	0.2940
B_xi_11	gas	0.3843	0.0375	0.3109	0.4578
B_xi_12	gas	0.5194	0.0435	0.4343	0.6046
B_xi_13	gas	0.3143	0.0777	0.1619	0.4666
gamma	positive_static	0.0905	0.0334	0.0249	0.1560
zeta	positive_static	6.0518	0.2389	5.5835	6.5202
omega0	pi	-0.5848	0.5270	-1.6178	0.4482
rho	pi	0.0225	0.1219	-0.2165	0.2615
omega_y_1	pi	0.0083	0.0070	-0.0053	0.0220
omega_y_2	pi	0.0075	0.0082	-0.0087	0.0237
omega_y_3	pi	0.0027	0.0030	-0.0032	0.0086
omega_y_11	pi	0.0187	0.0072	0.0046	0.0329
omega_y_12	pi	0.0114	0.0164	-0.0208	0.0436
omega_y_13	pi	0.0078	0.0083	-0.0084	0.0241

Estimated Parameters — Monthly ϕ, ξ

Note: Estimates from the optimizer used for the cached model. Standard errors and 95 % Wald CIs from the inverse Hessian; interpret with caution near optimizer bounds.

Model Comparison Tables

Section Overview

This section compares all models within each cadence (daily, monthly) side by side. Separate tables are provided for IS and OOS samples. The best value per metric column is highlighted in green. Red cells in coverage tables indicate rejection at the 5 % level.

Daily Models

Performance metrics and coverage tests for the daily cadence models.

Performance Metrics — Daily Models (IS)

Model	loglik	aic	bic	rmse	mad	crps
Daily ϕ -only	-5035.184	10126.367	10297.087	12.085	7.513	2.678
Daily ϕ, ξ	-5017.619	10121.238	10383.415	13.313	8.921	2.667

Note: Best value per metric highlighted in green. Lower is better for CRPS, RMSE, MAD, AIC, BIC; higher is better for log-likelihood.

Performance Metrics — Daily Models (OOS)

Model	rmse	mad	crps
Daily ϕ -only	11.638	7.575	2.620
Daily ϕ, ξ	13.669	9.162	2.631

Note: Best value per metric highlighted in green. Lower is better for CRPS, RMSE, and MAD.

95 % Coverage Tests — Daily (IS)

Model	violations	expected_violations	violation_rate	LR_uc	pvalue_uc	LR_cc	pvalue_cc	reject_uc_5pct	reject_cc_5pct
Daily ϕ -only	139.000	164.250	0.042	4.301	0.038	5.870	0.053	1.000	0.000
Daily ϕ, ξ	143.000	164.250	0.044	3.020	0.082	9.310	0.010	0.000	1.000

Note: Kupiec LR_uc tests unconditional 95 % coverage; Christoffersen LR_cc adds independence of violations. Red cells indicate rejection at the 5 % significance level.

95 % Coverage Tests — Daily (OOS)

Model	violations	expected_violations	violation_rate	LR_uc	pvalue_uc	LR_cc	pvalue_cc	reject_uc_5pct	reject_cc_5pct
Daily ϕ -only	34.000	36.550	0.047	0.192	0.662	0.461	0.794	0.000	0.000
Daily ϕ, ξ	32.000	36.550	0.044	0.621	0.431	0.761	0.684	0.000	0.000

Note: Kupiec LR_uc tests unconditional 95 % coverage; Christoffersen LR_cc adds independence of violations. Red cells indicate rejection at the 5 % significance level.

95 % Coverage Tests — Daily (Full)

Model	violations	expected_violations	violation_rate	LR_uc	pvalue_uc	LR_cc	pvalue_cc	reject_uc_5pct	reject_cc_5pct
Daily ϕ -only	173.000	200.800	0.043	4.242	0.039	5.107	0.078	1.000	0.000
Daily ϕ, ξ	175.000	200.800	0.044	3.641	0.056	8.403	0.015	0.000	1.000

Note: Kupiec LR_uc tests unconditional 95 % coverage; Christoffersen LR_cc adds independence of violations. Red cells indicate rejection at the 5 % significance level.

Monthly Models

Performance metrics and coverage tests for the monthly cadence models.

Performance Metrics — Monthly Models (IS)

Model	loglik	aic	bic	rmse	mad	crps
Monthly ϕ -only	-1053.809	2157.619	2240.449	197.350	129.046	29.047
Monthly ϕ, ξ	-1017.678	2111.357	2237.259	226.185	160.741	22.986

Note: Best value per metric highlighted in green. Lower is better for CRPS, RMSE, MAD, AIC, BIC; higher is better for log-likelihood.

Performance Metrics — Monthly Models (OOS)

Model	rmse	mad	crps
Monthly ϕ -only	183.041	126.160	26.753
Monthly ϕ, ξ	579.005	312.083	92.704

Note: Best value per metric highlighted in green. Lower is better for CRPS, RMSE, and MAD.

95 % Coverage Tests — Monthly (IS)

Model	violations	expected_violations	violation_rate	LR_uc	pvalue_uc	LR_cc	pvalue_cc	reject_uc_5pct	reject_cc_5pct
Monthly ϕ -only	6.000	10.150	0.030	2.080	0.149	2.448	0.294	0.000	0.000
Monthly ϕ, ξ	11.000	10.150	0.054	0.073	0.787	1.341	0.512	0.000	0.000

Note: Kupiec LR_uc tests unconditional 95 % coverage; Christoffersen LR_cc adds independence of violations. Red cells indicate rejection at the 5 % significance level.

95 % Coverage Tests — Monthly (OOS)

Model	violations	expected_violations	violation_rate	LR_uc	pvalue_uc	LR_cc	pvalue_cc	reject_uc_5pct	reject_cc_5pct
Monthly ϕ -only	4.000	1.200	0.167	4.391	0.036	4.572	0.102	1.000	0.000
Monthly ϕ, ξ	6.000	1.200	0.250	10.803	0.001	11.202	0.004	1.000	1.000

Note: Kupiec LR_uc tests unconditional 95 % coverage; Christoffersen LR_cc adds independence of violations. Red cells indicate rejection at the 5 % significance level.

95 % Coverage Tests — Monthly (Full)

Model	violations	expected_violations	violation_rate	LR_uc	pvalue_uc	LR_cc	pvalue_cc	reject_uc_5pct	reject_cc_5pct
Monthly ϕ -only	10.000	11.350	0.044	0.176	0.675	0.759	0.684	0.000	0.000
Monthly ϕ, ξ	17.000	11.350	0.075	2.585	0.108	2.661	0.264	0.000	0.000

Note: Kupiec LR_uc tests unconditional 95 % coverage; Christoffersen LR_cc adds independence of violations. Red cells indicate rejection at the 5 % significance level.

Conclusion

Summary of Findings

This report presents the first systematic comparison of two ZA-GAS specifications on the Belo Horizonte precipitation time series at daily and monthly cadences.

- Daily OOS — lowest CRPS: Daily ϕ -only (2.6199).
- Daily OOS — lowest RMSE: Daily ϕ -only (11.6384).
- Monthly OOS — lowest CRPS: Monthly ϕ -only (26.7528).
- Monthly OOS — lowest RMSE: Monthly ϕ -only (183.0407).

Coverage Diagnostics

Full-sample 95 % coverage is rejected for: Daily ϕ -only, Daily ϕ , ξ .

Calibration (PIT)

PIT uniformity rejected at 5 % for: Monthly ϕ -only.

Next Steps

- Extend the analysis to all Brazilian weather stations to assess spatial consistency.
- Compare ZA-GAS against benchmark models (climatological quantiles, seasonal ARIMA) on the same evaluation metrics.
- Investigate whether ξ dynamics consistently improve calibration in high-rainfall regimes.
- Explore alternative lag sets or seasonal parameterisations.

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